

POST-GROKING COLLAPSE AT THE REPRESENTATION-READOUT INTERFACE IN MUON-TRAINED TRANSFORMERS

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Paper under double-blind review

ABSTRACT

Under the standard routing that assigns Muon the hidden weight matrices and AdamW the embeddings and output head, Muon reaches the grokking threshold on modular arithmetic in fewer steps. Every configuration in our original Muon sweep subsequently loses generalization, with training accuracy failing alongside test; AdamW at larger step sizes collapses too. Freezing the interface suppresses collapse in matched branches. At five captured transitions, the readout update is sufficient for the acute loss of accuracy, while fresh linear decoders recover 98.20–100% from collapsed representations. The measured pairs fail tests of exact global basis equivalence. Correcting cross-entropy arithmetic prevents joint train/test collapse through step 30,000 in all five matched branches; the original failing updates locally descend the accurate loss and overshoot at their applied scale. An RMS-normalized counterexample instead fails through its embeddings despite accurate arithmetic. Fourier filtering separates failure through the current readout from *circuit masking*, where the task-aligned family solves the task perfectly in isolation and is outvoted by the remainder; rescaling the family restores accuracy. At a captured collapse, dominant-frequency membership is unchanged and the power distribution remains nearly identical. These spectral statistics alone do not establish that a learned circuit still functions.

1 INTRODUCTION

Mechanistic accounts of grokking rest on two progress measures: the set of Fourier frequencies a circuit occupies and the fraction of representational power they carry (Nanda et al., 2023). Together they are the standard certificate that a circuit is present. We exhibit a regime in which the certificate and the function come apart. At the step a grokked transformer stops computing modular addition, the dominant-frequency membership of the task-aligned family is unchanged and its power distribution retains a cosine similarity of 0.9899 (Figure 1). This spectral stability raises a natural hypothesis: the representation has changed basis while the readout has failed to compensate. We test that hypothesis directly. Fresh decoders recover the task, but measured residuals fail exact global basis-equivalence tests. Preserved task information and preserved spectral statistics therefore require separate tests of the mechanism.

The vehicle is Muon (Jordan et al., 2024), used as deployed at scale (Liu et al., 2025): hidden weight matrices receive orthogonalized momentum, embeddings and the output head remain with AdamW. Under this routing Muon reaches the grokking threshold on modular arithmetic faster than AdamW (Tveit et al., 2025; Wang, 2026), and we reproduce the result. The solutions it reaches do not hold. Every configuration in our stock-arithmetic sweep on $(a + b) \bmod 113$ groks and every one subsequently loses generalization, repeatedly, over hundreds of thousands of steps. Training accuracy falls with test accuracy in the same evaluation interval, so the failure is not a loss of generalization over a solved training set. AdamW at larger step sizes also collapses, with severity differing across optimizers. The instability persists across moduli, widths, training fractions, both operations, and depths 1, 2, and 4. The speedup is itself the instrument: it places a grokked circuit within a few thousand steps, which lets us observe the post-grokking regime repeatedly inside single runs, branch matched trajectories from bit-identical parameter states, and intervene on the circuit at a cost that is prohibitive at AdamW’s timescale.

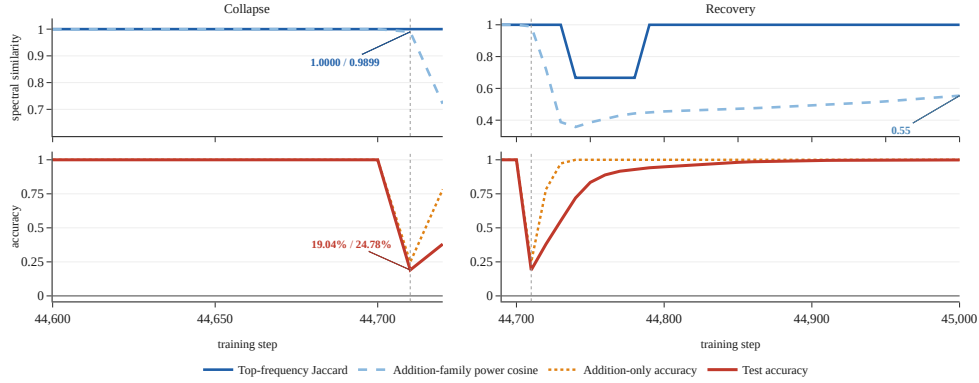


Figure 1: A model and its filtered pathway failing, at ten-step evaluation resolution. **Left:** across a single marked interval, test accuracy falls from 100% to 19.04% and accuracy restricted to the addition family to 24.78%, while within the family the set of dominant frequencies is unchanged (Jaccard 1.0000) and the power distribution across them retains cosine similarity 0.9899. These spectral statistics remain stable despite the functional failure. **Right:** the model returns to 99.90% within 300 steps with the power cosine to the pre-collapse state reaching only 0.55, having re-solved the task on a different set of frequencies. The addition family is the non-constant (k, k) modes of the two-dimensional Fourier transform of the final residual over the operand grid.

The unnormalized failures localize to the representation–readout interface. The logits depend on the final residual h and the unembedding W_U through $W_U h$, so $(Rh, W_U R^{-1})$ implements the same function for any invertible R (Elhage et al., 2021). This parameterization symmetry alone does not specify an optimizer trajectory. Branching from bit-identical states, freezing either side prevents the imminent failure over the tested window. Anchoring the embeddings and readout after circuit formation yields no sub-threshold evaluation across five runs totaling 451,400 post-grokking steps; five paired seeds also remain stable while their unfrozen arms record 137 to 321 such evaluations each. Adjacent-checkpoint interventions then isolate the acute readout update, and fresh decoders establish that substantial task information survives it.

Filtering the representation to the Fourier family the task selects separates two failure modes. In *circuit failure*, the filtered family fails through the current readout; this is a pathway-level measurement, compatible with recovery by a fresh decoder. In *circuit masking* it solves the task perfectly while the full model does not: read through the unembedding, the family casts a correct vote on every example, larger on the examples the model answers wrongly, and is outvoted by the rest of the representation. Rescaling the family alone, with nothing retrained, restores the model. In the measured AdamW trajectory, grokking is the same competition resolving upward: the family solves the task in isolation long before the model does, and accuracy tracks its share of representational power.

Numerical controls connect these events to prior work on softmax collapse and finite-precision loss spikes (Prieto et al., 2025; Liu Hanqing et al., 2026). Correcting cross-entropy from step 6,000 prevents joint train/test collapse through step 30,000 in five matched branches. The captured failing updates are local descent directions for the accurate loss, but their full applied steps overshoot. Gain-free RMS normalization exposes a boundary: an accurate-loss run still collapses through an embedding update. The numerical contribution and the acute locus therefore depend on the architecture.

Contributions.

- **Phenomenon.** Post-grokking collapse occurs throughout the original Muon sweep and under AdamW at larger step sizes, with joint failures of training and test accuracy (§3).
- **Mechanism.** Matched freezes, adjacent-update interventions, fresh decoders, and arithmetic corrections identify an acute readout failure with surviving task information in the unnormalized model, and an embedding failure in an RMS control (§4).

Table 1: Parameter groups and routing. The AdamW baseline places all parameters under one AdamW instance. Selected settings: Muon hidden $\text{lr} = 0.03$, $\text{wd} = 0.1$, $\beta = 0.95$, five Newton–Schulz iterations; auxiliary $\text{lr} = 10^{-3}$, $\text{wd} = 1.0$; readout $\text{lr} = 2.5 \times 10^{-4}$, $\text{wd} = 1.0$; AdamW baseline $\text{lr} = 10^{-3}$, $\text{wd} = 3.0$. Selection protocol and full sweeps in Appendix B.

Group	Parameters	Optimizer
Hidden	QKV, attention output, MLP input, MLP output, per block	Muon
Auxiliary	Token embedding, position embedding	AdamW
Readout	Unembedding	AdamW

- **Taxonomy.** Filtering to the task-selected Fourier family separates circuit failure from circuit masking; masking is a competition on amplitude that rescaling alone resolves, with the reverse transition observed during grokking (§6).
- **Methodology.** Stable spectral statistics can coexist with functional failure. Decoder and basis tests distinguish retained information from exact coordinate equivalence (Figure 1, §6, Appendix J.2).

2 SETUP

Task. We train on $(a + b) \bmod p$ with $p = 113$ throughout, and on $(a - b) \bmod p$ where stated, on the token sequence $[a, b, =]$ with 113 output classes. A fixed random 30% of the p^2 operand grid is the training set, trained full batch with cross-entropy; the original experiments evaluate every 100 steps. The arithmetic follow-ups use the additional monitoring in §4.4.

Architecture. The model is a decoder-only transformer, $d_{\text{model}} = 128$, four heads, MLP width 512, ReLU, sequence length 3, every projection bias-free, and no normalization layers. An invertible change of residual basis can be absorbed by the matrices that write to and read from it. This is a symmetry of the parameterization; optimizer updates need not preserve it. The gain-free RMS control restricts this symmetry to $O(d)$. Depth is varied by appending blocks under nested initialization.

Routing and definitions. Parameters split into the three groups of Table 1; the hidden group is the only one Muon receives. *Sustained 95%* is the first evaluation followed by five at or above 95% test accuracy, and grokking speed is always measured against it; *strictly stable* means no post-grokking evaluation below 95%, and we report the count and depth of sub-threshold evaluations alongside the label. Matched branches fork from a single in-memory state with zero parameter difference; seed replications run on a bitwise-deterministic backend with the two arms verified identical at every logged evaluation before they are made to differ (Appendix A).

3 MUON GROKS FASTER, AND THE SOLUTIONS DO NOT HOLD

All nine Muon configurations in a 3×3 sweep of the hidden group reach sustained generalization, against seven of eleven AdamW configurations, with mean times of 13,011 and 30,486 steps; the mean is computed over AdamW’s successes alone, so the factor of 2.34 is conservative. On the selected settings the advantage is 1.54 at the median over five seeds, and Muon’s grokking step spans one hundred steps where AdamW’s spans seven thousand seven hundred. Under a smaller modulus, a smaller training fraction, and half width the ratios grow to 4.9, and with depth past 5.7 (§7).

None of the nine Muon configurations is strictly stable: sub-threshold counts range from 2 to 592, post-grokking minima reach 0.78%, and the counts have no monotone relationship to either hidden hyperparameter. Four of the seven grokking AdamW configurations also collapse, with severity rising monotonically in the learning rate, from none at 10^{-3} to 918 sub-threshold evaluations at 10^{-2} . Sweeping the readout learning rate over a factor of five and its weight decay from zero to 0.5 changes the timing and depth of collapse and eliminates none of it; the zero-decay case removes the weight-decay pull on the readout and leaves the failure in place. Across five seeds neither selected setting is strictly stable: Muon records 137 to 321 sub-threshold evaluations on every seed with minima to 16.27%, and the AdamW reference records one or two on four of five, with a minimum of

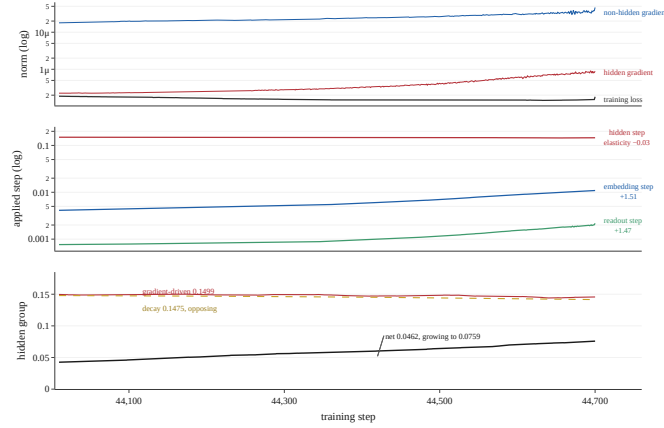


Figure 2: The 691 steps before a collapse, at every step. **Top:** the training loss falls while both gradient norms rise. **Middle:** the gradient-driven applied step of each group, with the log–log elasticity on its own gradient norm: Muon’s does not track the gradient, the two AdamW groups track it more than proportionally. **Bottom:** on the hidden group the gradient-driven step and the weight decay oppose each other to within two percent, so the net displacement is a small residual, and that residual grows by 64% while the step producing it stays flat.

27.59%. The instability transfers to modular subtraction with the same asymmetry. Both optimizers collapse, with different severity. Full sweeps, the baseline-selection protocol, and generality variants are in Appendix B.

4 LOCALIZING THE COLLAPSE AT THE INTERFACE

4.1 THE GRADIENT VANISHES AND THE HIDDEN STEP DOES NOT

Long before a collapse the model has solved the training set: over the 691 steps preceding the event we analyze, the training loss sits at 1.5×10^{-7} and the gradient norms on the hidden and non-hidden groups are of order 10^{-6} and 10^{-5} (Figure 2). The Newton–Schulz iteration normalizes its input before orthogonalizing and returns singular values near one whatever produced it, so a decoupled step is the premise of the update rule; the window quantifies it and what follows from it. The elasticity of each group’s gradient-driven applied step on its own gradient norm, the slope of a log–log regression across the window, is -0.03 for the hidden group against $+1.5$ for the embeddings and the readout, with R^2 between 0.84 and 0.98. AdamW’s scale invariance assumes a constant gradient rescaling; here the magnitude drifts over hundreds of steps. On the hidden group the orthogonalized step and the weight decay oppose each other to within two percent, and the net displacement is the residual, which grows by 64% across the window while the step producing it stays flat. The consequence is the finding: the hidden group accumulates 0.073 of displacement per parameter in root mean square against 0.0091 for the non-hidden group, a factor of 8.0, and the instantaneous ratio swings five-fold across the window, which no fixed ratio of learning rates produces. These stock-arithmetic measurements establish different displacement scales (Appendix C); their causal interpretation requires the numerical controls in §4.4. The failure then occupies a single evaluation interval: the training loss rises by a factor of 3.8×10^8 , and training and test accuracy fall together, from 100% to 21.12% and 19.04%. Nothing about the orthogonalized step is anomalous at the moment the model breaks.

4.2 WHICH PARAMETERS THE FAILURE REQUIRES

Table 2 reports branches from a common state, each run for 2,000 steps with a different subset of parameters held fixed. Freezing either group suppresses the failure for the whole window; freezing the token or position embedding delays it without preventing it; freezing the unembedding prevents it over the same window, and leaving only the hidden matrices and the unembedding free reproduces

Table 2: Matched branches from a single state at step 44,000, each run to step 46,000. Collapse is recorded when training accuracy falls below 90%. Freezing either optimizer-managed side of the interface prevents it over this window; within the auxiliary group only the unembedding does, and hidden matrices together with the unembedding reproduce it.

Branch	First train collapse	Min test	Final test	Max aux. grad.
<i>Group branches</i>				
Control	44,700	16.77%	100.00%	913.07
Freeze hidden	none	99.07%	99.08%	2.79×10^{-5}
Freeze auxiliary	none	100.00%	100.00%	0
<i>Auxiliary component branches</i>				
Control	44,710	43.57%	99.93%	329.29
Freeze token embedding	44,910	34.74%	99.90%	3278.60
Freeze position embedding	44,760	37.78%	99.92%	323.76
Freeze unembedding	none	100.00%	100.00%	4.94×10^{-6}
Hidden and unembedding only	45,290	40.80%	100.00%	348.00

it. With the auxiliary group frozen, Muon continues to apply hidden updates of 0.146 and hidden displacement nearly doubles while the loss holds at 1.5×10^{-7} .

These branches establish suppression over a finite window; freezing also changes subsequent gradients and optimizer history. To isolate the failing update, Figure 4a swaps features and heads between adjacent checkpoints at five captured events. In every seed, the old head retains or improves accuracy on the updated features, while the new head sharply lowers accuracy on the old features. Fresh linear decoders fitted using only training examples recover 98.20–100% at selected collapsed checkpoints (Appendix J.2). Longer branches still show coadaptation: each decodes its own representation and fails on the other’s, and initially identical heads reach cosine similarity -0.028 . These observations establish readout sensitivity and surviving information. Exact global basis equivalence fails for the measured pairs, despite successful recovery of planted transformations. The joint normalization/orthogonalization ablation also has a separate limitation: all six generalizing runs end in non-finite loss (Appendix D).

4.3 ANCHORING THE INTERFACE PREVENTS THE COLLAPSE

We hold the token embedding, the position embedding, and the unembedding constant from roughly two thousand steps after sustained generalization, while hidden matrices continue to receive Muon updates; we call the configuration *Stable Muon*. Across five runs spanning the main condition, $p = 97$, training fraction 20%, and width 64, a total of 451,400 post-grokking steps and 4,519 evaluations, no evaluation falls below 95%, and in every run the post-grokking minimum is the accuracy at the evaluation that first met the criterion. Figure 3 pairs the arms at five seeds: freezing removes every sub-threshold evaluation on every seed, against an AdamW reference that achieves strict stability on one seed of five. Grokking speed is preserved because the freeze follows the threshold, and per-step cost falls afterward. In the long component comparison, holding only the unembedding fixed leaves eighteen of 936 evaluations below threshold; also freezing the embeddings removes them (Appendix E). Arithmetic correction supplies another successful intervention over a shorter horizon.

4.4 NUMERICAL CONTROLS AND THE FAILING STEP

Five matched branches replace stock cross-entropy with a cancellation-resistant `loglp` evaluation from step 6,000, after grokking and measurable derivative error. Through step 30,000, none has a joint train/test failure below 90%; minimum measured test accuracy is 98.86%, and all finish at 100%. Every-update training checks trigger test evaluation below threshold, detecting every joint event. Brief test-only excursions can fall between grid evaluations. From step 15,000 to 20,000, accurate loss, target-derivative repair, and per-example zero-sum gradient projection each give 0/5 failures (Appendix J.3). Projection retains substantial derivative error while restoring the class-sum identity, connecting the result to numerical feature inflation (Liu Hanqing et al., 2026).

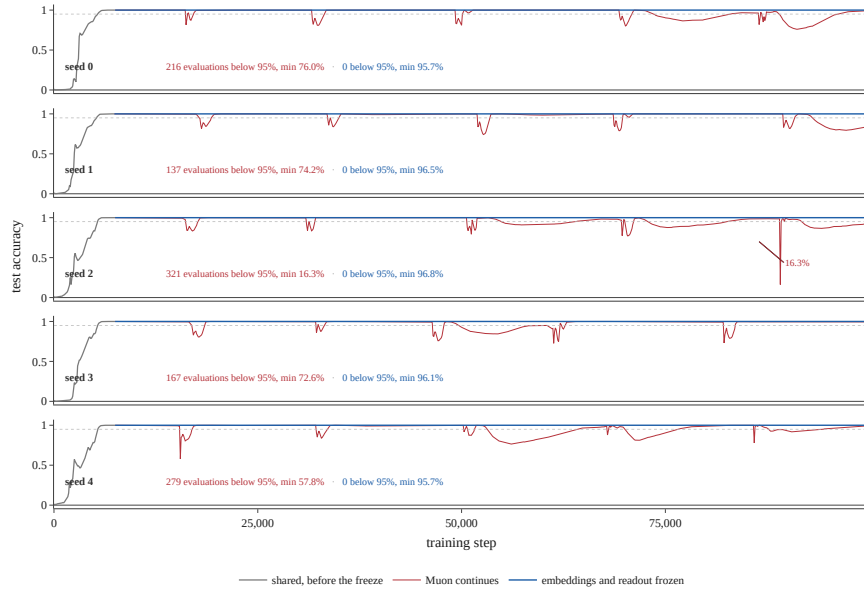


Figure 3: The intervention at five seeds. Each trajectory is shared until the freeze, with the two arms identical to a maximum absolute difference of zero across every logged evaluation, and then continues once under ordinary Muon and once with the embeddings and unembedding held constant. The unfrozen arm falls below threshold 137 to 321 times per seed with minima as low as 16.27%; the frozen arm records none and never falls below 95.67%.

Before the captured failures, full-grid feature-mean norms reach 10,374–12,203. An intervention using the following representation’s training mean nearly reproduces each failure with only the mean-dependent part of the readout update. Replacing gradients only at the terminal update still fails in all five seeds, implicating accumulated state. Each complete update has negative directional derivative under the accurate training loss, and scales 0.001 and 0.01 lower that loss; the full update overshoots (Appendix J.4).

One matched subtraction pair reproduces prevention through 30,000 steps. A gain-free RMS pair bounds its scope: with accurate loss, test accuracy falls from 96.21% to 0.88%. Updating only embeddings gives 0.94%, while updating only the readout retains 96.21%. Accurate arithmetic therefore leaves a distinct embedding failure possible; this one-seed counterexample does not estimate its prevalence (Appendix J.5).

5 THE TASK SELECTS THE FAMILY; THE UPDATE RULE SHAPES ITS SPREAD

Every analysis below views the model’s internal state as a function of the operand pair. Running the model on all p^2 pairs and applying an orthonormal two-dimensional Fourier transform over the operand axes decomposes any cached state into modes (k, l) , which partition into six families: *addition* (k, k) , *subtraction* $(k, -k)$, *a-only*, *b-only*, the constant mode, and generic interaction. *Sufficiency* of a family is the accuracy when the state is filtered to it; *necessity* is the accuracy when it is zeroed and everything else retained. The families are disjoint and exhaustive, so the two are exact complements. The full intervention suite is specified in Appendix F.

One family computes the task, and the task chooses it. Table 3: the addition family reaches exactly 100.00% in isolation at every solved checkpoint, with no variation by seed, optimizer, or training step, and removing it leaves near chance while removing any other family costs almost nothing. Trained on $(a - b) \bmod 113$ under matched initialization, configuration, and code path, the two families exchange roles exactly: $(k, -k)$ is sufficient everywhere and (k, k) sits at chance to the digit. Nothing in the analysis privileges either family; the task selects it, extending across optimizer choice the universality Chughtai et al. (2023) report across architectures and seeds.

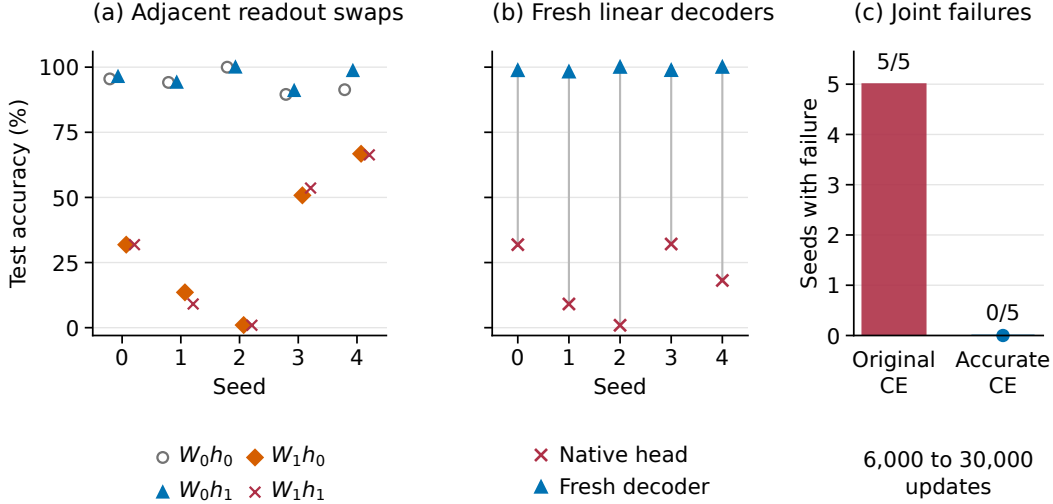


Figure 4: Follow-ups on five selected unnormalized trajectories. **(a)** Adjacent swaps: subscripts 0 and 1 denote states before and after the captured update. The old head retains accuracy on updated features; the new head sharply lowers accuracy on old features. **(b)** Training-only decoders recover 98.20–100% at collapsed snapshots; seeds 3 and 4 use later snapshots than (a). **(c)** Every original trajectory has a captured joint train/test failure below 90%; accurate CE from the matched step-6,000 state yields 0/5 through 30,000. Panels reuse seeds. Counts report binary incidence under mixed original monitoring and complete joint-event detection in corrected branches (Appendix J).

Table 3: Family sufficiency and ablation on the two operations. Addition columns at depths 2 and 4 aggregate eleven solved checkpoints; depth-1 columns are matched to the subtraction runs on initialization, configuration, and code path. Chance is $1/113 = 0.885\%$; across forty-three solved checkpoints spanning five seeds and three regimes, the task-aligned family gives exactly 100.00% at every one (Appendix F).

	Addition, depths 2 and 4	Addition, depth 1	Subtraction, depth 1
(k, k) sufficiency	100.00%	100.00%	0.88%
$(k, -k)$ sufficiency	0.88%	0.88%	100.00%
(k, k) ablation	1.34%	1.51%	99.68%
$(k, -k)$ ablation	98.26%	97.51%	2.54%
a -only / b -only / constant suff.	0.88%	0.88%	0.88%
Generic sufficiency	1.38%	1.59%	2.38%
Solved checkpoints	11	6	4

Frequency and phase matter within the family. Transporting the family onto other diagonal frequencies at matched power gives 2.64%. Randomizing phase per conjugate pair gives 0.19%, below chance: a coherent phase rotation leaves the model systematically wrong, which is what an algorithm reading the answer off a sum of angles produces when the angles are offset. Randomizing phase per channel gives 0.78%. Two branches diverging from a bit-identical state each decode their own representation at 100% and the other’s at chance, in both directions: the model relies on particular frequencies, particular phases, and the readout that grew beside them.

The update rule shapes the spread. On matched runs the Muon representation occupies 326 effective conjugate pairs of the non-constant spectrum against AdamW’s 5, leaving the task-aligned family holding 28% of non-constant power against AdamW’s 91%, and the contrast replicates across seeds and inside the family itself (Figure 5; Appendix G). Prior evidence for this difference is spectral (Ruan et al., 2026; Wang et al., 2025; Wang, 2026); measuring it in the basis the model computes in reads the spread against the family carrying the algorithm. Ablating the normalization and orthogonalization together collapses the count to 4, past AdamW’s, while leaving the family untouched at

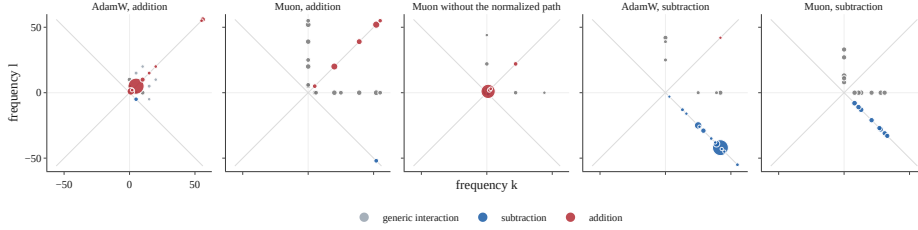


Figure 5: The twenty highest-power conjugate pairs of the final residual at converged checkpoints, on the mode grid, sized by share of non-constant power; grey lines mark $l = k$ and $l = -k$. AdamW places 80.1% of non-constant power on a single pair of the family the task selects; Muon’s strongest pair carries 7.9%, with the rest spread across the family and beyond it; ablating the normalized path returns the strongest pair to 58.9%. Between the addition and subtraction panels the occupied line moves from one diagonal to the other.

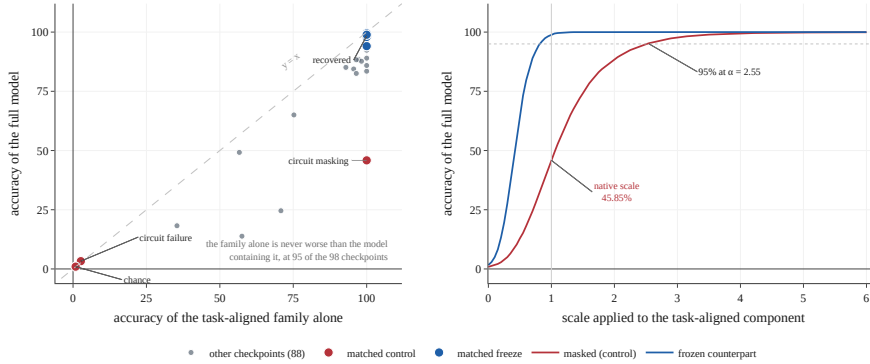


Figure 6: **Left:** task-aligned family accuracy against full-model accuracy over 98 checkpoints from the original studies, both evaluated through the current readout. *Circuit failure* denotes failure of this filtered pathway; *circuit masking* sits at the lower right, where the family solves the task and the model does not. At 95 of 98 checkpoints the family alone is at least as accurate as the full model, and the three exceptions exceed it by under a point. **Right:** rescaling the task-aligned component of the masked checkpoint by α , with the rest of the representation and the readout untouched.

100% sufficiency: the task selects the family, the update rule shapes how widely power spreads, and the two are separable.

6 A CIRCUIT CAN BE PRESENT, CORRECT, AND OUTVOTED

Filtering the final residual to the task-aligned family and decoding measures its accuracy through the current readout; comparison with the unfiltered model measures interference from the remainder. The two measurements separate two failures (Figure 6). In *circuit failure*, filtering gives 2.65% and then chance: this filtered pathway fails, and the family has fallen out of the top twenty power pairs almost entirely. Fresh-decoder recovery in §4.2 shows why such failure alone cannot establish loss of task information. In *circuit masking*, filtering gives 100% while the full model reaches 45.85%: the isolated circuit is perfect, and the representation it sits in overwhelms it. One matched trajectory passes through failure, full recovery, and masking; its frozen counterpart holds pairs, share, and accuracy throughout (Appendix H).

Masking is a competition on amplitude. Because the unembedding is linear and bias-free, the correct-class margin decomposes exactly into the task-aligned contribution and the remainder (Table 4). The family produces a positive margin on every example in both branches, including every example the masked model answers wrongly, and on those examples its margin is higher, +6.08

Table 4: Correct-class margin through the unembedding at the masked checkpoint, decomposed exactly by linearity of the bias-free readout; the competitor class is fixed per example as the runner-up under the full model. The final column is the fraction of examples on which the component alone is positive.

Branch	Component	Mean	On correct	On wrong	Fraction > 0
Control	full	-0.71	2.15	-3.13	0.4585
Control	task-aligned	+5.72	5.30	6.08	1.0000
Control	remainder	-6.43	-3.15	-9.21	0.0475
Freeze	full	+11.61	11.76	-1.67	0.9889
Freeze	task-aligned	+19.28	19.30	17.18	1.0000
Freeze	remainder	-7.67	-7.54	-18.86	0.0622

against +5.30. The failures coincide with stronger opposition from the remainder: the remainder contributes negatively on more than nine tenths of examples in both branches, and in the control an opposition of -6.43 outvotes a task-aligned term of +5.72. Rescaling the task-aligned component by α , with nothing retrained, raises the masked model monotonically from 45.85% to 95% at $\alpha = 2.55$ and 99.91% at $\alpha = 6$: the operative quantity is the ratio of the family’s margin to the opposition, and the family’s share of representational power is its descriptive proxy.

Grokking as masking resolving upward. Along an AdamW trajectory tracked at thousand-step resolution, the addition family is sufficient at 95.14% at step 3,000, when the model scores 0.30% and the family holds 1.05% of non-constant power; its complement never computes the task at any point in training. Which algorithm the representation contains is settled long before generalization. What changes is the share, rising to 70.62% at the test-accuracy jump, and the concentration within the family. The accuracy transition follows the rising share: the cleanup phase of Nanda et al. (2023) is this quantity rising, circuit-masking collapse is the same quantity falling, and the two transitions are one measurement run in opposite directions. Neither requires weight decay on the readout, which the zero-decay sweep of §3 removes.

What the standard measures report. Exact nonzero Fourier support is invariant under an invertible residual-basis map; power and its rankings are invariant under orthogonal maps. General invertible maps can change dominant-frequency membership. A joint, compensated basis change also preserves predictions. At the collapse in Figure 1, the measured dominant-frequency set has Jaccard 1.0000 and power cosine 0.9899, while the isolated pathway’s mean margin moves from +18.14 to -46.85. Over this ten-step interval the unembedding moves by 1.285 of its norm and the representation by 0.392; the old head on new features scores 26.31%, against 5.74% for the reverse. Adjacent-update swaps provide finer localization (Figure 4a). Recovery changes the frequency allocation: accuracy returns to 99.90% within 300 steps, while the old representation remains poorly decoded by the recovered head over the measured window. Similarly, the frozen branch’s addition component scores 99.99% through its own head and 5.05% through the control’s. These cross-readout failures show coadaptation; exact basis drift requires the separate tests in Appendix J.2.

7 CONSTRUCTION AND ALIGNMENT SEPARATE IN DEPTH

Depth grows the speed advantage, from $2.22\times$ at depth 2 to past $5.7\times$ at depth 4, where the AdamW baseline never sustains generalization, and it removes the condition that separated the optimizers: the AdamW configuration stable at depth 1 records 556 sub-threshold evaluations at depth 2. Stage-level interventions locate the code. Frequency- and phase-specific sensitivity first appears at an MLP activation in ten of eleven solved checkpoints and at no attention write, while every earlier stage is computationally necessary without carrying the code, which is why ablation alone cannot locate a circuit. Decoding a post-block residual directly with the unembedding reaches 95% only at the final block in all 41 generalizing checkpoints and never exceeds 17.97% earlier: an MLP in the penultimate block writes the code and the final block makes it legible, so adding depth adds stages whose residual states are not aligned with the readout and does not distribute the alignment across them. Compatibility with the final readout is concentrated at the last block; anchoring the interface at depth 4 cuts sub-90% evaluations twenty-fold on an exact-state matched branch (Appendix I).

8 RELATED WORK

Grokking on modular arithmetic. Power et al. (2022) introduced the phenomenon; Nanda et al. (2023) reverse-engineered the Fourier algorithm and the progress measures this paper tests, establishing that the circuit forms before the accuracy jump and that the jump is a cleanup phase. Zhong et al. (2023) show hyperparameters select qualitatively different algorithms; Chughtai et al. (2023) establish universality across architectures and seeds.

Failure after grokking. Thilak et al. (2022) report cyclic instability under adaptive optimizers; Varma et al. (2023) describe ungrokking under reduced data; Prieto et al. (2025) identify softmax collapse and a loss-minimization direction that scales logits. Liu Hanqing et al. (2026) connect finite-precision violations of the logit-gradient class-sum identity to feature inflation and loss spikes, and test projection-based corrections. Our arithmetic controls build on this account; adjacent swaps and fresh decoders locate the acute functional failure and surviving information. Joint train/test collapse differs from the perfect-training regime of anti-grokking (Prakash & Martin, 2025), while the accurate-loss RMS event establishes a separate architectural boundary.

Muon. Jordan et al. (2024) introduced the optimizer and Liu et al. (2025) deployed it at scale under the routing used here; Tveit et al. (2025) and Wang (2026) report faster grokking, the latter isolating orthogonalization as the cause and five Newton–Schulz iterations as the robust choice over short horizons; the collapse here occurs at those five iterations over longer ones. Ruan et al. (2026) and Wang et al. (2025) report higher effective rank and a more isotropic spectrum under Muon; we measure the same difference in the basis the model computes in and implicate the normalized-orthogonalization path by ablation.

Representation and readout. Elhage et al. (2021) state the residual stream’s basis freedom, which motivates our explicit equivalence tests. Anthes et al. (2023) identify readout misalignment as a major component of forgetting under new tasks; Chou et al. (2026) and Gomezjurado Gonzalez (2026) decompose the approach to grokking into representation and readout speeds. Here the data and task stay fixed, and the dissociation occurs after generalization. Jointly failing training and test predictions can coexist with high accuracy under a fresh decoder.

9 DISCUSSION

The tests establish surviving task information, with training-only decoders reaching 98.20–100%, while rejecting exact global basis equivalence for the measured pairs. Arithmetic corrections suppress joint failures over fixed horizons; captured updates overshoot accurate local descent directions. Together with adjacent swaps and the feature-mean intervention, these results support a numerical contribution to acute readout failure. A unique mediator of the accumulated trajectory remains unresolved.

The accurate-loss RMS embedding failure limits this account. The main controls cover five selected trajectories; the operation and normalization extensions use one seed each. Longer horizons and brief test-only excursions remain unresolved, and AdamW collapses limit attribution specifically to Muon. For interpretability, spectral stability, linear decodability, and accuracy through the current readout are distinct measurements. Assessing whether a learned circuit still works requires all three.

REPRODUCIBILITY STATEMENT

The original experiments run on a single device from released code; their repository (anonymized for review) contains the training and analysis scripts, 191 CSV run logs, and 20 HTML summaries. Follow-up artifacts add raw diagnostics, model/optimizer/random-state checkpoints, replay checks, and the scripts for every new table and figure. Matched freezes and arithmetic branches verify shared states before intervention. Appendix A specifies the original setup; Appendix J gives the decoder, basis, arithmetic, and architecture-control protocols and their fixed horizons.

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Table 5: Operational definitions. Evaluations occur every 100 steps, so a sustained criterion requires the threshold to hold across a 500-step window.

Term	Definition
Sustained 99.9% train	First evaluation followed by five evaluations at or above 99.9% train accuracy
Sustained 95% test	First evaluation followed by five evaluations at or above 95% test accuracy
Memorization plateau	Sustained-95 test step minus sustained-99.9 train step
Strictly stable	No post-grokking evaluation below 95% test accuracy

Table 6: Selected configurations. Stable Muon and Muon are the same configuration up to the freeze; their trajectories differ only through backend nondeterminism, which Section 4.3 reports.

Configuration	Group	Optimizer	Settings
AdamW baseline	All parameters	AdamW	$\text{lr} = 10^{-3}$, $\text{wd} = 3.0$
Muon	Hidden	Muon	$\text{lr} = 0.03$, $\text{wd} = 0.1$, $\beta = 0.95$, $\text{NS} = 5$
	Auxiliary	AdamW	$\text{lr} = 10^{-3}$, $\text{wd} = 1.0$
	Readout	AdamW	$\text{lr} = 2.5 \times 10^{-4}$, $\text{wd} = 1.0$
Stable Muon	Muon, with auxiliary and readout groups frozen after circuit formation		

A SETUP DETAILS AND OPERATIONAL DEFINITIONS

Each example is the token sequence $[a, b, =]$; the vocabulary contains $p + 1 = 114$ tokens and the output layer $p = 113$ classes. The ordered operand grid contains $p^2 = 12,769$ examples; a fixed random 30% (unless stated) forms the training set, generated once from the run seed and shared by all matched comparisons. The original runs load a saved initial state from disk, evaluate every 100 steps, and checkpoint every 1,000. Depth-1 and depth-2 runs last 100,000 steps and depth-4 runs 300,000.

Each block holds four two-dimensional matrices (fused QKV 384×128 , attention output 128×128 , MLP input 512×128 , MLP output 128×512), 196,608 parameters; with embeddings and unembedding the depth-1 model has 226,048 parameters. Depth is varied by appending blocks under nested initialization, so deeper models begin from a strict superset of the shallower model’s initial state. Muon maintains an exponential moving average of the gradient with $\beta = 0.95$, forms a Nesterov-style update, normalizes it by its Frobenius norm, applies five quintic Newton–Schulz iterations with coefficients $(3.4445, -4.7750, 2.0315)$, scales by $\sqrt{\max(1, \text{rows}/\text{cols})}$, and applies decoupled weight decay; the implementation refuses any parameter that is not a matrix. All AdamW groups use $\beta_1 = 0.9$, $\beta_2 = 0.999$.

Table 5 gives the operational definitions for the original experiments; evaluations occur every 100 steps, so a sustained criterion holds across a 500-step window. Training sweeps and matched-branch experiments run at a single seed on an accelerator backend that is not bitwise deterministic across processes; the five-seed replications run on CPU, where execution is bitwise deterministic, so the Muon and Stable Muon arms are the same trajectory until the freeze fires and agree to a maximum absolute difference of zero over the seventy-four to seventy-five evaluations preceding it. The seed sets the initialization and the train/test split together, so the five seeds are five draws of the problem instance.

B SWEEPS, BASELINE SELECTION, AND GENERALITY

Tables 7 and 8 give both sweeps under identical architecture, split, and budget. The fastest AdamW configuration ($\text{lr} = 10^{-2}$, $\text{wd} = 1.0$) is unusable: 918 sub-threshold evaluations at this seed, and across seeds three of five never reach sustained 95%, the two that do record 805 and 956, and all five end within a tenth of a point of chance. The baseline is therefore the fastest configuration that is strictly stable in this sweep, $\text{lr} = 10^{-3}$, $\text{wd} = 3.0$; the seed replication shows strict stability at that setting is a property of one seed rather than of the configuration. For Muon no configuration is strictly stable, so the selected configuration is the fastest.

Table 7: AdamW sweep. Seven of eleven retained configurations reach sustained 95% test accuracy. Post-grokking evaluations below 95% increase monotonically with the learning rate.

LR	WD	Sustained 95%	Min after grokking	Evals < 95%	Strictly stable
3×10^{-4}	0.1	no grok			
3×10^{-4}	1.0	no grok			
3×10^{-4}	3.0	25,500	95.67%	0	yes
10^{-3}	0.1	no grok			
10^{-3}	1.0	52,500	96.49%	0	yes
10^{-3}	3.0	8,200	95.01%	0	yes
3×10^{-3}	0.1	49,500	13.26%	2	no
3×10^{-3}	1.0	8,500	53.72%	4	no
10^{-2}	0.1	62,900	0.54%	258	no
10^{-2}	1.0	6,300	0.41%	918	no
10^{-2}	3.0	no grok			

Table 8: Muon sweep over the hidden group. All nine configurations reach sustained 95% test accuracy and none is strictly stable. The count of sub-threshold evaluations has no monotone relationship to either hyperparameter.

LR	WD	Sustained 95%	Min after grokking	Evals < 95%	Strictly stable
0.03	0.1	5,400	69.40%	56	no
0.01	0.1	6,000	53.24%	107	no
0.03	0.3	7,200	75.63%	2	no
0.01	0.3	8,500	76.60%	145	no
0.03	0.03	9,000	0.78%	592	no
0.003	0.1	14,400	90.07%	5	no
0.003	0.03	19,000	21.24%	130	no
0.01	0.03	19,500	0.83%	473	no
0.003	0.3	28,100	78.59%	265	no

Both optimizers fit the training set between steps 200 and 1,400 in every grokking configuration, so the difference lies entirely in the delay from memorization to generalization: 30,100 steps for AdamW against 12,544 for Muon averaged over successes. Across five seeds Muon reaches sustained generalization between 5,300 and 5,400 steps and the AdamW reference between 4,000 and 11,700, medians 5,400 and 8,300.

Varying the readout learning rate over 10^{-4} , 2.5×10^{-4} , and 5×10^{-4} gives post-grokking minima of 37.13%, 79.40%, and 3.81%; varying its weight decay over 0, 0.1, and 0.5 gives 33.34%, 0.79%, and 1.12%; every setting is unstable. On $(a - b) \bmod 113$ the Muon run records 18 post-grokking evaluations below 95% with a minimum of 1.52% and the AdamW run records none, reproducing the depth-1 addition asymmetry at 194 and none. Table 9 varies modulus, training fraction, and width; Stable Muon and Muon share a configuration up to the freeze, so differences in their grokking steps bound run-to-run variation on this backend.

C THE QUIET WINDOW

Table 10 follows the 691 steps preceding the collapse of §4.1 and the failure interval itself. On the hidden group the gradient-driven component is 0.1499 at step 44,100 against a decay component of 0.1475, a ratio of 0.984, leaving a net displacement of 0.0462 that grows to 0.0759 across the window while the orthogonalized step stays flat; over the window the hidden gradient norm grows by 3.92 while its gradient-driven step changes by 0.97, and the non-hidden gradient grows by 2.60 while the embedding and readout steps grow by 2.65 and 2.84. The readout is pulled by decay rather than by the loss on 14.9% of quiet-window steps. The instantaneous per-parameter displacement ratio between the groups falls from 16.2 to 3.3 across the window. At the failure the readout gradient norm rises from 3.52×10^{-5} to 836.10; the orthogonalized Muon step over that interval is 0.1429, lower than the 0.1455 preceding it, and its maximum over the branch occurs after the event.

Table 9: Controlled variation of modulus, training fraction, and width. Stable Muon and Muon share a configuration up to the freeze, so differences in their grokking step measure run-to-run variation on this backend rather than an effect of the intervention.

Variant	Regime	Sustained 95%	Min post-grokking	Outcome
$p = 97$, train 30%, width 128	AdamW	31,000	95.52%	stable
	Muon	6,300	15.49%	unstable
	Stable Muon	6,300	95.05%	stable
$p = 113$, train 20%, width 128	AdamW	no grok		
	Muon	15,100	49.25%	unstable
	Stable Muon	14,500	95.16%	stable
$p = 113$, train 30%, width 64	AdamW	84,800	96.38%	stable
	Muon	17,700	53.23%	unstable
	Stable Muon	16,500	95.67%	stable

Table 10: Diagnostics through the quiet window and across the collapse. The training loss and both gradient norms are numerically negligible until the failure. The Muon update column is the orthogonalized step before weight decay, constant to within three percent; the net displacement after decay is smaller and grows across the window, and is reported in the text.

Step	Train loss	Hidden grad. norm	Non-hidden grad. norm	Muon update norm
44,100	1.77×10^{-7}	2.33×10^{-7}	1.95×10^{-5}	0.1498
44,300	1.55×10^{-7}	2.73×10^{-7}	2.20×10^{-5}	0.1494
44,500	1.50×10^{-7}	4.06×10^{-7}	2.67×10^{-5}	0.1481
44,700	1.53×10^{-7}	8.64×10^{-7}	3.52×10^{-5}	0.1455
44,710	57.50	25.26	836.10	0.1429
44,720	6.58	5.09	92.55	0.1655

D ABLATING THE NORMALIZED-ORTHOGONALIZATION PATH

The ablation removes the Frobenius normalization and the Newton–Schulz iteration together, leaving momentum, shape scaling, and weight decay; the update carries the magnitude and direction of the buffer, and the learning rate must rise to 0.3 or 1.0 for the model to learn at all. Table 11: six ablated runs generalize, none records a post-grokking evaluation below 95%, and all six end in a non-finite loss, in float64 as well as float32, at terminal steps spanning 22,889 to 38,955 and at hidden norms that set no failure threshold. Muon on the matched run reaches a minimum of 29.49% and recovers to 96.48%, with thirteen sub-threshold evaluations before the earliest ablated failure: the ablated runs span the window in which Muon oscillates and none of them does. The two failures differ in kind, and the ablation collapses spectral dispersion from 326.09 effective conjugate pairs to 4.11 while leaving the task-aligned family at 100% sufficiency (Appendix G).

E PREVENTION DETAILS

In four of the five runs the freeze is triggered automatically, 2,000 steps after the first evaluation of a five-evaluation streak at or above 95%; the main-condition run uses a fixed step. Across all five the freeze lands 2,100 to 2,200 steps after sustained generalization. Within a depth-2 run the median step costs 57.87 ms before the freeze and 54.72 after; at depth 4, 111.39 and 107.75; pre-freeze rates match ordinary Muon’s. Holding only the unembedding constant from the same step leaves the run unstable, minimum 83.13% and eighteen of 936 evaluations below 95%; freezing the embeddings and unembedding together at the same step produces none.

F FOURIER DECOMPOSITION AND INTERVENTION SUITE

For a chosen point in the forward pass the model runs on all p^2 operand pairs, the states form a tensor $X \in \mathbb{R}^{p \times p \times d}$, and an orthonormal two-dimensional DFT over the operand axes gives $\hat{X}[k, l, :]$ with

Table 11: Ablated runs that reach sustained generalization. Minimum is the lowest test accuracy from the sustained-95 step to the last recorded evaluation. Terminal is the step at which the loss becomes non-finite.

Learning rate	Backend	Precision	Grokked	Minimum	Terminal
0.3	MPS	float32	3,100	97.28%	23,514
0.3	CPU	float32	3,100	97.52%	24,485
0.3	CPU	float64	3,100	97.06%	22,889
1.0	MPS	float32	6,800	97.11%	23,800
1.0	CPU	float64	7,700	95.96%	29,831
1.0	CPU	float32	6,500	95.60%	38,955

Table 12: Stable Muon across five runs covering four settings; the main condition appears twice, under a fixed freeze step and under the automatic trigger. The minimum is the lowest test accuracy at any evaluation from the sustained-95 step onward, and in every run it is the value at that step itself.

Run	Grokked	Frozen	Post-grokking steps	Minimum	Evals < 95%
Main condition	5,900	8,100	94,100	95.34%	0
Selected Stable Muon	5,400	7,500	94,600	95.74%	0
$p = 97$	6,300	8,400	93,700	95.05%	0
Training fraction 20%	14,500	16,600	85,500	95.16%	0
Width 64	16,500	18,600	83,500	95.67%	0
Total			451,400		0

power $\|\hat{X}[k, l, :]\|^2$. Because X is real, modes occur in conjugate pairs and every filter retains or removes both members together; transforms run in single precision on CPU. The families of Table 13 are pairwise disjoint and exhaust the grid, and addition and subtraction are disjoint because p is odd. Power fractions are normalized against non-constant power; filters that retain the addition family retain the full diagonal including the constant mode, since the bias-free unembedding turns a constant component into a fixed logit offset. Removing a family and retaining its complement are the same measurement, and they agree exactly everywhere, checking the filters.

Table 15 applies sufficiency and ablation to each family at forty-three solved checkpoints spanning five seeds and the Muon, Stable Muon, and AdamW regimes. Equal-power relocation ranges 0.88% to 15.04% over 430 replicates. Taken in descending order of power, reaching 95% requires two to four conjugate pairs at AdamW checkpoints (median 3), against three to seventeen at Muon (median 12.5) and eight to nineteen at Stable Muon (median 11). Along the AdamW trajectory of §6, the five strongest pairs give 4.91% at step 3,000, 19.48% at 20,000, 70.73% at 30,000, and 100% at 37,000, while the family’s share rises from 1.05% through 11.08% and 33.20% to 70.62%, reaching 90.51% by step 100,000; test accuracy over those steps is 0.30%, 2.08%, 15.97%, and 100%.

G SPECTRAL DISPERSION

Table 16 reports the inverse participation ratio over all 6,384 non-constant conjugate pairs and the task-aligned family’s power share on matched depth-1 runs. Across the five-seed study AdamW occupies 4.40 effective pairs on average over thirty solved checkpoints, Muon 385.85 over twenty-five, and Stable Muon 343.96 over thirty-six: freezing the interface leaves the dispersion where Muon put it and removes the instability anyway. Within the family’s 56 pairs, AdamW occupies 1.99 to 2.59 effective pairs across the generality variants and Muon 21.33 to 55.34. Each operation supplies a family that provably computes nothing on the other’s task: AdamW places 0.0–0.3% of non-constant power there and Muon 2.5–17.7%.

H THE MATCHED BRANCH THROUGH FAILURE, RECOVERY, AND MASKING

Table 17 follows the family’s top-twenty membership, power share, isolated accuracy, and full-model accuracy along the depth-4 matched pair of §6 from a bit-identical state. The frozen branch

Table 13: The six mode families. The families are pairwise disjoint and exhaust the $p \times p$ grid, so any statement about one family and its complement is an exact accounting of the whole representation.

Family	Modes	Count at $p = 113$
Addition	$(k, k), k \neq 0$	112
Subtraction	$(k, -k), k \neq 0$	112
a -only	$(k, 0), k \neq 0$	112
b -only	$(0, l), l \neq 0$	112
Constant	$(0, 0)$	1
Generic interaction	all remaining (k, l)	12,320
Total		12,769

Table 14: Interventions applied to a cached state. Each is designed to hold one property of the representation fixed while destroying another, so that a drop in accuracy attributes to the property destroyed.

Intervention	Preserved	Destroyed
Family sufficiency	Modes in one family	All other families
Family ablation	All other families	Modes in one family
Equal-power relocation	Total power, family membership	Which frequencies carry it
Random native subset	Family membership, power scale	Which of the used frequencies survive
Pair-global phase scramble	Support and power exactly	Relative phase across pairs
Channelwise phase scramble	Support and power exactly	Relative phase across channels
Cross-readout substitution	Both representations intact	Pairing of state with its own readout

holds all three quantities for 174,000 steps; the control loses all three at filtered-pathway failure through its current readout, recovers all three, and loses the share again at circuit masking while the isolated family still gives 100%. The α -sweep crossing at $\alpha = 2.55$ occurs at a share of 0.6247, below the frozen branch’s native 0.9069. Replaying a depth-1 collapse and recovery at ten-step resolution: full accuracy returns to 99.90% within 300 steps while the addition-family power cosine to the pre-collapse state reaches only 0.55; the pre-collapse representation holds between 4.2% and 5.7% under the recovered readout throughout, while the recovered representation reads at 90.73% under the pre-collapse readout. The matched branches’ five strongest modes are identical at the branch point and share none by steps 160,000 through 230,000. Substituting the frozen branch’s addition component into the control under the frozen readout gives 99.99%; under the control’s own readout, 5.05%; the frozen branch’s full final representation gives 97.61% and 4.46% under the two readouts; substituting only the frozen branch’s non-addition component into the control gives 36.79%, below the control’s unmodified 45.85%.

I DEPTH

Muon reaches sustained generalization in 2,300 steps at depth 2 against 5,100, and in 52,600 at depth 4 where the baseline peaks at 96.69% and never sustains the threshold within 300,000 steps; Table 18 gives the comparison in steps and elapsed time, with a Muon step costing $1.75\text{--}1.80\times$ an AdamW step. Table 19: zeroing the block-2 or block-3 attention write leaves 0.94% and 0.44%, while relocating the family or scrambling its phase at those stages costs nothing; sensitivity appears abruptly at the block-3 MLP activation and every later stage is at or below chance. The depth-2 checkpoint behaves identically with the transition at the block-1 MLP activation. Across the eleven solved checkpoints of the depth sweep the first stage where pair-global phase scrambling costs more than half the accuracy is an MLP activation or write in ten, and an attention write in none. Table 20: direct decoding reaches 95% only at the final block across all 41 generalizing checkpoints, never exceeding 17.97% earlier; filtering to the family raises accuracy at sixteen of seventeen partially aligned residual states, by 4.88 points on average, so the competition of \$6 is visible along the forward pass and not only across training. Comparing the depth-4 matched arms at every checkpoint after their shared state, the power spectral cosine between them stays above 0.98 in the first block and falls to 0.03–0.44 in the fourth at fourteen of nineteen: the branches part company in the block that feeds the readout. The exact-state matched branch at step 126,200 raises mean post-branch

Table 15: Family interventions at forty-three solved checkpoints, spanning five seeds and the Muon, Stable Muon, and AdamW regimes. Sufficiency retains only the named family; ablation removes it and retains the rest. The chance rate is 0.88%.

Family	Sufficiency	Range	Ablation	Range
Addition	100.00%	[100.00, 100.00]	1.71%	[0.67, 4.24]
Subtraction	0.88%	[0.88, 0.88]	99.61%	[95.79, 100.00]
<i>a</i> -only	0.88%	[0.88, 0.88]	99.85%	[97.76, 100.00]
<i>b</i> -only	0.88%	[0.88, 0.88]	99.85%	[97.72, 100.00]
Constant	0.88%	[0.88, 0.88]		
Generic interaction	3.28%	[0.78, 14.21]	99.99%	[99.78, 100.00]

Table 16: Spectral dispersion at solved checkpoints of the matched depth-1 runs. Effective pairs is the inverse participation ratio over all 6,384 non-constant conjugate pairs, reported as a mean with range. Family power is the task-aligned family’s share of non-constant power, meaning (k, k) for addition and $(k, -k)$ for subtraction, and top-five power is the share carried by the five strongest pairs of the spectrum. The final row ablates the normalization and orthogonalization together.

Operation	Optimizer	Effective pairs	Family power	Top-five power
Addition	AdamW	4.95 [3.68, 10.20]	91.0%	93.7%
Addition	Muon	326.09 [96.91, 578.90]	28.0%	13.1%
Subtraction	AdamW	2.54 [2.45, 2.72]	95.4%	96.5%
Subtraction	Muon	211.46 [33.25, 377.24]	27.7%	16.0%
Addition	Muon, path ablated	4.11 [2.04, 6.05]	66.5%	94.0%

accuracy from 48.07% to 97.50%, final accuracy from 36.88% to 98.87%, and cuts evaluations below 90% from 1,353 to 66.

Table 17: The addition family along matched branches. Pairs counts how many of the twenty highest-power conjugate pairs at the final residual belong to the family; share is the family’s fraction of non-constant power; addition-only is accuracy when the representation is filtered to the family.

Step	Branch	Pairs	Share	Addition-only	Full
126,200	control	11	83.95%	100.00%	94.07%
126,200	freeze	11	83.95%	100.00%	94.07%
160,000	control	1	11.88%	2.65%	3.29%
160,000	freeze	12	89.33%	100.00%	98.97%
180,000	control	0	0.32%	0.88%	0.97%
180,000	freeze	12	88.32%	100.00%	97.95%
230,000	control	4	67.69%	100.00%	98.41%
230,000	freeze	12	90.78%	100.00%	99.47%
300,000	control	3	20.38%	100.00%	45.85%
300,000	freeze	12	90.69%	100.00%	98.89%

Table 18: Time to sustained generalization, in optimizer steps and in elapsed seconds on a single device. Per-step cost is the median over evaluation intervals. The AdamW baseline at depth 4 is right-censored at the step budget.

Depth	Optimizer	ms/step	Steps to grok	Seconds to grok	Advantage
2	AdamW	33.06	5,100	170	2.22× steps, 1.28× time
	Muon	57.81	2,300	133	
4	AdamW	61.68	> 300,000	> 19,768	> 5.70× steps, > 3.37× time
	Muon	110.87	52,600	5,856	

Table 19: Stage-level interventions at depth 4, step 53,000, baseline 99.30%. Relocation transports the addition family onto other diagonal frequencies at matched power; phase scrambling randomizes the phase of each conjugate pair; zeroing removes the component entirely.

Stage	Kind	Relocation	Phase	Zeroed
Block 1 attention write	write	99.30%	99.30%	86.31%
Block 2 attention write	write	99.33%	99.29%	0.94%
Block 3 attention write	write	99.29%	99.30%	0.44%
Block 3 MLP activation	activation	1.66%	0.00%	1.19%
Block 3 MLP write	write	0.95%	0.00%	1.19%
Block 4 attention write	write	3.58%	0.00%	58.44%

Table 20: Direct-readout accuracy by post-block residual, decoding with the unembedding and skipping the remaining blocks. Values are ranges over checkpoints at which the model generalizes.

Depth	Regime	Earlier blocks	Final block
2	AdamW	0.7–1.1%	100.00%
2	Muon	0.8–5.1%	95.8–100.00%
2	Stable Muon	1.3–5.2%	96.4–99.4%
4	Muon	0.8–18.0%	95.1–99.3%
4	Stable Muon	0.7–18.1%	95.2–98.7%

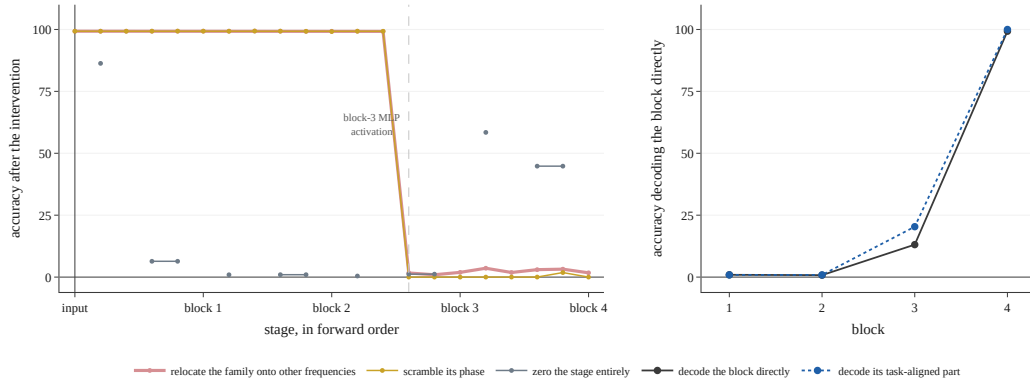


Figure 7: A depth-4 Muon checkpoint. Left: accuracy after intervening on each stage in forward order. Relocating the task-aligned family onto other frequencies and scrambling its phase leave the model above 99% at every stage through the block-3 attention write and destroy it at the block-3 MLP activation; the two coincide before the cliff. Zeroing a stage destroys accuracy everywhere, so earlier stages are computationally necessary while carrying no frequency- and phase-specific code. Right: decoding a post-block residual directly with the unembedding, for the state itself and for its task-aligned part alone. Neither is legible before the final block, and the task-aligned part is the more legible of the two wherever both are partial.

J DECODER, BASIS, AND ARITHMETIC FOLLOW-UPS

The decoder and matched-branch tests use the original architecture, data splits, and optimizer settings; operation and normalization controls are specified separately below. The basis study adds 52,698 baseline updates and 9,000 continuation updates. The subsequent causal study adds 364,302 updates: five 24,000-step arithmetic branches, fifteen 5,000-step specificity branches, four original-trajectory extensions totaling 49,302 steps, and four 30,000-step operation or architecture controls. Verification replays are excluded from these totals. Multiple checkpoints and interventions from one trajectory are dependent observations.

J.1 MONITORING AND MATCHED STATES

The causal branches and fresh baseline replications use a joint event: training and test accuracy must both fall below 90% after sustained grokking. Training accuracy is checked at every update; test accuracy is checked every 100 updates and at every post-grokking state with training accuracy below 90%. Thus every joint event is detected, while brief test-only excursions can fall between evaluations. Six consecutive grid evaluations at or above 95% confirm grokking. The confirmation step is the *last* evaluation of that streak, 500 updates after the streak-start definition used for grokking speed in §2.

Within each seed, matched arithmetic branches restore identical parameters, optimizer buffers, and random-number-generator state. Exact next-update replays reproduce all three. The original prefixes combine historical evaluation grids with densely captured failures; their positive events establish binary incidence. We therefore compare event presence, without comparing their counts, durations, or historical minima against the uniformly monitored branches. The five trajectories were selected for mechanistic study and give no estimate of collapse prevalence over an unrestricted seed population.

Table 21: Test accuracy (%) at adjacent captured failures. h_0, W_0 precede the update; h_1, W_1 follow it. Each head is evaluated with each feature state, without retraining. Original events are captured under mixed historical monitoring grids (Appendix J.1).

Seed	Before: $W_0 h_0$	Old head: $W_0 h_1$	New head: $W_1 h_0$	After: $W_1 h_1$
0	95.49	96.39	31.86	31.87
1	94.14	94.18	13.56	9.09
2	99.99	100.00	1.01	0.93
3	89.54	91.01	50.80	53.60
4	91.34	98.66	66.75	66.33

J.2 FRESH DECODERS AND THE BASIS HYPOTHESIS

Fresh bias-free linear decoders are fitted to the raw final residual using only the original 3,830 training examples. An uncentered SVD preconditioner retains all 128 dimensions, with singular values floored at 10^{-6} of the largest. Each inner fit estimates its own preconditioner. A 20% split of the training set selects the L_2 penalty from $\{0, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$ by validation cross-entropy; the selected decoder is then refitted on all training examples. The 8,939 test examples enter only the final evaluation. Fits use float64 logistic regression and L-BFGS-B; convergence and unregularized controls are recorded.

At the selected collapsed checkpoints, decoder accuracies for seeds 0–4 are 98.72%, 98.20%, 100.00%, 98.77%, and 100.00%, against 99.51%, 99.56%, 100.00%, 99.40%, and 100.00% at their references. Probe checkpoints are steps 17,494, 17,721, 18,069, 16,309, and 16,060: the last two occur after the adjacent transitions in Table 21. These results establish substantial surviving linear task information; the small gaps in three seeds leave room for reduced decodability or probe limitations.

For column residuals, an exact global basis account requires one invertible R satisfying $h_1(x) = R h_0(x)$ across the operand grid. With examples stacked as rows, full column rank makes this equivalent to equality of the two feature column spaces. We test train-fitted linear and affine maps, centered reconstruction, rank sensitivity, principal angles, and a separately identified full-domain

oracle. Centering by training means prevents a large constant feature component from dominating reconstruction error.

Table 22: Centered held-out reconstruction error (%) for a train-fitted map from later to earlier residuals. Healthy controls match interval length. Reference intervals span 6–508 updates and occur at different optimization stages, so comparisons are within each seed.

Seed	Reference to captured failure	Healthy interval	Adjacent failure update
0	43.63	32.66	1.88
1	43.05	10.82	4.63
2	9.05	2.76	1.10
3	45.73	14.30	10.66
4	25.60	1.18	25.63

Positive controls plant exact residual gauges into the trained network. Eight controls cover condition numbers 1, 3, 10, and 100 in float32 and float64; twenty-four further controls plant observed fitted matrices with their measured orientation and conditioning. Exact compensation and train-fitted compensation recover 100% in every fixed-condition control. For the three fresh adjacent transitions, observed forward-map errors are 4.65%, 1.10%, and 10.77%, against 0.0004–0.0006% for their planted controls. These errors reject exact global equivalence for the measured pairs. Healthy evolution also departs from exact equivalence (Table 22).

Approximate transport remains useful: inverse-linear compensation reaches 95.13%, 100%, and 94.37% at the three fresh first events. Reference timing matters. Seed 0’s head from step 17,200 scores 14.05% on the collapsed residual, while the head from 17,493 scores 96.39%. Its geometric discrepancy from the distant reference is already present before the final failing update. Moreover, the complete real task-only Fourier coefficient matrices have shape 113×128 ; when both have full row rank, an invertible map between them always exists. A fitted task-only map can therefore align that projection exactly while scoring only 4.62% on raw held-out features.

One-time repairs also have limited persistence. Eighteen continuations resume two captured failures for 500 updates under nine head/state interventions each. Every successful repair retaining optimizer state falls below 90% training accuracy on the next update. Clearing the head’s Adam state improves eventual recovery but permits the immediate relapse. These tests separate instantaneous decodability from stability under continued optimization.

J.3 ARITHMETIC INTERVENTIONS

The accurate loss evaluates $\log(1 + \sum_{j \neq y} \exp(z_j - z_y))$ with $\log 1p$ whenever the correct-class logit is maximal; other examples use ordinary cross-entropy. Model storage, backward matrix arithmetic, and Muon’s update remain float32. This avoids cancellation in the small loss and correct-class derivative, following the numerical concerns of Prieto et al. (2025). All five step-6,000 sources already satisfy the six-check grokking criterion and already contain measurable gradient error. The intervention acts on an existing post-grokking trajectory.

Table 23: Matched arithmetic branches from step 6,000 through 30,000. Original columns show a captured joint failure. Accurate-CE branches have no joint failure; all finish at 100% train and test accuracy. Minima include the starting state.

Seed	Original event step	Original event test (%)	Accurate minimum test (%)
0	17,494	31.872	99.754
1	17,721	9.095	99.843
2	18,069	0.929	98.859
3	16,308	53.597	99.978
4	16,056	66.327	99.843

The specificity study branches each seed at step 15,000 and runs to 20,000 with three corrections. Accurate CE gives 0/5 joint failures; target-derivative repair gives 0/5; per-example zero-sum projection gives 0/5. Every original trajectory has a captured failure in this window. All fifteen corrected branches finish at 100% test accuracy; the minimum measured test accuracy across these branches is 99.978%.

For stock logit gradient g , target repair retains each wrong-class entry and sets $g_y = -\sum_{j \neq y} g_j$, leaving the reported forward loss unchanged. Projection replaces g by $g - \mathbf{1}(\mathbf{1}^\top g)/p$. Derivative errors are relative L_2 errors against the accurate float64 derivative evaluated on the identical float32 logits. The readout-gradient reference casts this logit derivative to float32 before backpropagating through the shared float32 network. At the fixed seed-4, step-16,000 audit, target repair reduces relative derivative error from 32.02% to 4.19×10^{-7} , whereas projection leaves 31.88%. At step 20,000, projected derivatives retain 52.94–86.07% relative error despite a class-sum residual below 2.1×10^{-8} relative to the reference norm. Thus restoring the class-sum identity can prevent the observed failures over this window even with substantial remaining derivative error. Numerical feature inflation and this projection control build on Liu Hanqing et al. (2026).

At step 15,000, stock logit-gradient errors are 10.56–21.86% and readout-gradient errors 18.95–43.42%. Passing a class-common gradient error through the same saved Adam state produces a class-centered update difference: its centered norm fraction is 45.85–73.37%, versus approximately 3×10^{-16} for scalar SGD. These descriptive norm ratios are nonadditive. Accurate-CE branches also have smaller feature means: original mean norms at step 15,000 are 2.22–3.83 times their matched corrected values.

J.4 MEAN AMPLIFICATION AND FINITE-STEP OVERSHOOT

At the checkpoints immediately before the five original failures, feature means over the full operand grid have norms 10,374–12,203. The intervention below uses the following checkpoint’s mean over training examples. Let h_1 be the updated residual, $W_{U,0}$ the preceding readout, ΔW_U its actual displacement, and μ this training mean of h_1 . Starting from the rescued logits $W_{U,0}h_1$, the readout-induced change decomposes exactly as

$$\Delta z = \Delta W_U(h_1 - \mu) + \Delta W_U\mu. \quad (1)$$

Applying the training-estimated mean unchanged to test examples, the mean term alone gives 32.11%, 9.42%, 0.93%, 53.34%, and 66.07% accuracy, close to the full failing updates in Table 23. The centered term alone retains 95.26%, 92.48%, 100%, 87.20%, and 98.46%. Relative reconstruction error is below 3×10^{-16} . This evaluation-only intervention establishes local sufficiency of the product of feature mean and readout displacement. Feature growth and optimizer history remain coupled before that step.

Replacing gradients only at the failing update with accurate gradients preserves all five failures. The state accumulated beforehand matters. For each actual complete parameter displacement $\Delta\theta$, the accurate training gradient satisfies $\nabla L^\top \Delta\theta < 0$, including each group’s contribution. Scaling the displacement by 0.001 or 0.01 lowers training loss in every seed; taking the full step raises it sharply. Both interpolation endpoints reproduce saved parameters exactly. The captured updates locally descend and overshoot at finite scale.

J.5 SUBTRACTION AND NORMALIZED ARCHITECTURE

Four additional runs use seed 0 and a fixed 30,000-step horizon: a stock/accurate-CE pair on subtraction and a pair on addition with gain-free RMS normalization before attention, before the MLP, and before the readout ($\epsilon = 10^{-6}$). Parameter initialization and optimizer settings are matched, with no added parameters or hyperparameter search.

Subtraction reproduces acute readout sensitivity: updated features with the preceding head score 98.81%, the mean term alone 5.84%, and the centered term 98.76%. Accurate CE prevents the joint event through the horizon and finishes at 99.966% test accuracy.

RMS normalization changes the acute locus. In the accurate-CE event, test accuracy falls from 96.208% to 0.884%; its logit derivative has relative error 9.48×10^{-8} and no vanished correct-class entries. Updating only the head retains 96.208%, updating only embeddings gives 0.940%, and

Table 24: Operation and architecture controls. Confirmation is the end of the six-check grokking streak. First-event test accuracies and post-grokking minima are separate measurements; both RMS runs subsequently recover to 100%. Each row is one seed.

Condition	CE	Confirmation	First event	Event test (%)	Min. test (%)
Subtraction	Stock	9,300	15,281	5.851	5.851
Subtraction	Accurate	10,500	none	n/a	96.879
RMS addition	Stock	15,700	18,515	85.658	0.481
RMS addition	Accurate	19,900	28,495	0.884	0.526

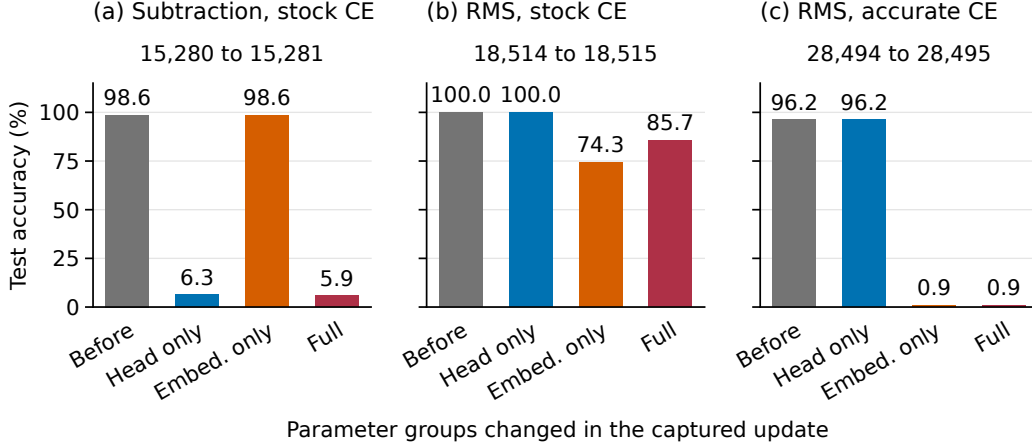


Figure 8: One-step parameter interventions at the first detected joint failure in each operation or architecture control, all at seed 0. Each bar uses the preceding checkpoint with only the indicated parameter group replaced by its following value; “Full” applies all three groups. In subtraction, the head update reproduces most of the damage. In both RMS cases, the embedding update is destructive while the head update preserves accuracy. The accurate-CE RMS failure occurs with an accurate loss derivative. These are first-event measurements at the displayed steps; the later post-grokking minima are reported separately in Table 24.

freezing the head still gives 0.895%. The minimum reaches 0.526% at step 28,503; the stock-RMS minimum is 0.481% at 28,540. Exact update replay passes. Both RMS event directions locally descend and lower loss at scales 0.001 and 0.01, then overshoot at full scale. This counterexample bounds the readout/feature-mean account to the measured unnormalized cases and rules out universal prevention by accurate loss arithmetic. The completed horizons support those finite claims; longer runs and population frequencies remain outside their scope.